

CHE 301 - Chemical Engineering Thermodynamics

Assignment 4

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Abstract

This report presents the thermodynamic analysis of two distinct problems involving real and ideal gas behavior. In **Problem A**, the real-gas properties of nitrogen were evaluated at $T = 60^\circ\text{C}$ and $P = 700$ bar using tabulated compressibility data. The density was interpolated through Akima spline fitting to obtain smooth, continuous temperature and pressure dependencies. From this, the isobaric thermal expansion coefficient, molar enthalpy, and molar entropy were determined by integrating the appropriate thermodynamic differential relations derived from the first and second laws. Ideal-gas reference properties and their corrections due to pressure effects were computed numerically in Excel.

In **Problem B**, the adiabatic and reversible venting of ethylene gas from a pressurized steel tank was analyzed to determine the mass of gas in the tank under both initial and critical conditions. The analysis employed conservation of mass, energy, and entropy under the assumption of ideal-gas behavior, accounting for the kinetic energy of the gas at the vent throat where sonic conditions occur. The report also examined how the tank mass changes as pressure decreases and related the results to choked flow behavior. Overall, the report demonstrates the application of thermodynamic principles and numerical interpolation methods to both real-gas and ideal-gas systems.

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1 Problem A: Statement

Given the tabulated values of $Z = PV/RT$ for Nitrogen gas at $T = \{0, 25, 50, 75, 100\}^{\circ}\text{C}$ and $P = \{0, 1, 3, 5, 10, 30, 50, 100, 200, 300, 400, 600, 800, 1000\}$ bar , compute at $T = 60.000^{\circ}\text{C}$ and $P = 700.000$ bar:

1. the thermal (volume) expansion coefficient in K^{-1} ,
2. the molar enthalpy in kJ/mol ,
3. the molar entropy in $\text{kJ}/(\text{kmol K})$.

	Temperature (deg C)				
P (atm)	0	25	50	75	100
0	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000
3	0.999	0.999	1.000	1.000	1.001
5	0.998	0.999	1.000	1.001	1.001
10	0.996	0.998	1.000	1.001	1.002
30	0.959	0.996	1.001	1.005	1.007
50	0.958	0.996	1.004	1.009	1.014
100	0.984	1.004	1.018	1.027	1.033
200	1.036	1.057	1.072	1.082	1.089
300	1.132	1.146	1.154	1.160	1.163
400	1.256	1.254	1.253	1.250	1.247
600	1.524	1.495	1.471	1.450	1.432
800	1.798	1.723	1.697	1.658	1.625
1000	2.067	1.988	1.923	1.867	1.819

Figure 1: Given compressibility data (PV/RT) for Nitrogen gas from Lunbeck, Michels, and Wolkers (1952).

1.1 Akima Spline and Excel Implementation

Definition of the Akima Spline

The Akima spline is a piecewise cubic interpolation technique developed to produce a smooth and physically consistent curve through a discrete set of data points. Unlike standard cubic splines, which can exhibit unrealistic oscillations between widely spaced points, the Akima method constructs local slopes using only neighboring data intervals. This localized approach allows each segment of the spline to depend primarily on its immediate surroundings, minimizing overshoot and ensuring continuity of both the function and its first derivative across the entire domain.

Mathematically, the Akima spline constructs a set of cubic polynomials between adjacent data points. For each interval $[x_i, x_{i+1}]$, the slope at each node is estimated using a weighted average of surrounding finite-difference gradients. This results in a curve that smoothly transitions through all data points while maintaining monotonicity where appropriate. The method is particularly

advantageous for experimental or tabulated thermodynamic data, where small deviations or noise in the measurements could otherwise lead to non-monotonic curvature.

In this problem, the Akima spline was implemented within Excel using the function:

`=fAkima(x_data, f_data, job, xv(optional), lower(optional), upper(optional))`

where:

- **x_data** represents the independent variable array,
- **f_data** represents the corresponding dependent variable array,
- **job** specifies the type of operation to perform,
- **xv** is the evaluation point,
- **lower** and **upper** define optional integration limits when **job** = 5.

The parameter **job** determines the operation mode according to:

job	Function
1	Interpolated value
2	Inverted value
3	First derivative
4	Second derivative
5	Definite integral

1.2 Applications in This Problem

The Akima spline was used for:

1. **Density interpolation at $T = 60^\circ\text{C}$:** Since tabulated data were only available at temperatures (0, 25, 50, 75, 100°C), the molar density at $T = 60^\circ\text{C}$ was determined for each pressure using Akima interpolation.

`=fAkima(T_range, rho_range, 1, 60)`

2. **Evaluation of $(\partial\rho/\partial T)_P$ at $P = 700$ bar:** The temperature derivative of density at constant pressure was obtained directly from the first derivative mode of the spline.

`=fAkima(T_range, rho_range, 3, 60)`

3. **Numerical integration of the H and S expressions:** The spline's integral mode (**job** = 5) was used to evaluate the pressure integrals appearing in the enthalpy and entropy relations:

$$H = H^{IG} + \int_0^P V(1 - \beta T) dP, \quad S = S^{IG} - \int_0^P \beta V dP.$$

1.3 Determination of Density

Relationship Between Z , V , and ρ

Starting from the definition of the compressibility factor:

$$Z = \frac{PV}{RT}, \quad (1)$$

the molar volume V and molar density ρ can be directly related as follows:

$$V = \frac{ZRT}{P}, \quad \rho = \frac{1}{V} = \frac{P}{ZRT}.$$

Hence, once Z is known at a given temperature and pressure, the molar density can be computed directly.

1.4 Density Calculation in Excel

The tabulated Z -data for Nitrogen were entered into Excel at the temperature points $T = \{0, 25, 50, 75, 100\}^\circ\text{C}$ and pressures $P = \{0, 1, 3, 5, 10, 30, 50, 100, 200, 300, 400, 600, 800, 1000\}$ bar. For each entry, the molar density was computed using the Excel function:

$$\text{=density}(T_C, P_bar, Z) \quad \text{which implements} \quad \rho = \frac{P}{ZRT}.$$

Density at $T = 60.000^\circ\text{C}$ Using Akima Spline Interpolation

Since Z (and therefore ρ) were not tabulated at $T = 60.000^\circ\text{C}$, the intermediate density values were obtained using an Akima spline interpolation between the five known temperatures (0–100 $^\circ\text{C}$) at each pressure.

In Excel, this was achieved using the Akima spline function:

$$\text{=fAkima}(T_C_range, \rho_kmol/m^3_range, 1, 60_C).$$

	Temperature, C					T, C
P, bar	0	25	50	75	100	Density, kmol/m ³
0	0	0	0	0	0	0
1.01325	0.044615033	0.04087	0.03771	0.035	0.03266	0.036584117
3.03975	0.1338451	0.12262	0.11314	0.10501	0.09798	0.109752352
5.06625	0.223075167	0.20437	0.18856	0.17502	0.16329	0.182920587
10.1325	0.446150334	0.40874	0.37712	0.35004	0.32659	0.365841173
30.3975	1.338451002	1.22622	1.13136	1.05012	0.97976	1.097523519
50.6625	2.23075167	2.0437	1.88559	1.75019	1.63294	1.829205865
101.325	4.461503341	4.0874	3.77119	3.50039	3.26587	3.658411731
202.65	8.923006681	8.17481	7.54238	7.00077	6.53174	7.316823461
303.975	13.38451002	12.2622	11.3136	10.5012	9.79761	10.97523519
405.3	17.84601336	16.3496	15.0848	14.0015	13.0635	14.63364692
607.95	26.76902004	24.5244	22.6271	21.0023	19.5952	21.95047038
810.6	35.69202672	32.6992	30.1695	28.0031	26.127	29.26729385
1013.25	44.61503341	40.874	37.7119	35.0039	32.6587	36.58411731

Figure 2: Interpolated and tabulated density data for Nitrogen at various temperatures and pressures.

1.5 Determination of the Thermal Expansion Coefficient

The thermal expansion coefficient, β , characterizes the temperature dependence of a substance's molar volume at constant pressure. It is defined as

$$\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P, \quad (2)$$

where V is the molar volume of the fluid.

Since the molar density ρ is the reciprocal of molar volume,

$$V = \frac{1}{\rho},$$

the derivative in Equation (2) can be expressed in terms of ρ as follows:

$$\left(\frac{\partial V}{\partial T} \right)_P = \frac{\partial}{\partial T} \left(\frac{1}{\rho} \right)_P = -\frac{1}{\rho^2} \left(\frac{\partial \rho}{\partial T} \right)_P. \quad (3)$$

Substituting this expression into Equation (2) yields the equivalent form:

$$\beta = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_P. \quad (4)$$

To compute $\left(\frac{\partial \rho}{\partial T} \right)_P$, the Akima spline interpolation was applied across the temperature range for each pressure. Using the Excel function:

`=fAkima(T_C_range, rho_kmol/m^3_range, 3, 60_C)`

the first derivative of density with respect to temperature was obtained at $T = 60^\circ\text{C}$ for each pressure value. The calculated slope $\left(\frac{\partial \rho}{\partial T} \right)_P$ was then substituted into Equation (4) to determine the corresponding β values.

						T,C	T,K	
	Temperature, C					60	333.15	
P, bar	0	25	50	75	100	Density, kmol/m3	(drho/dT), kmol/m3K	beta, 1/K
0	0	0	0	0	0	0	0	0
1.01325	0.044615033	0.04087	0.03771	0.035	0.03266	0.036584117	-0.000110063	0.003008
3.03975	0.1338451	0.12262	0.11314	0.10501	0.09798	0.109752352	-0.000330188	0.003008
5.06625	0.223075167	0.20437	0.18856	0.17502	0.16329	0.182920587	-0.000550313	0.003008
10.1325	0.446150334	0.40874	0.37712	0.35004	0.32659	0.365841173	-0.001100626	0.003008
30.3975	1.338451002	1.22622	1.13136	1.05012	0.97976	1.097523519	-0.003301878	0.003008
50.6625	2.23075167	2.0437	1.88559	1.75019	1.63294	1.829205865	-0.00550313	0.003008
101.325	4.461503341	4.0874	3.77119	3.50039	3.26587	3.658411731	-0.01100626	0.003008
202.65	8.923006681	8.17481	7.54238	7.00077	6.53174	7.316823461	-0.02201252	0.003008
303.975	13.38451002	12.2622	11.3136	10.5012	9.79761	10.97523519	-0.03301878	0.003008
405.3	17.84601336	16.3496	15.0848	14.0015	13.0635	14.63364692	-0.04402504	0.003008
607.95	26.76902004	24.5244	22.6271	21.0023	19.5952	21.95047038	-0.066037559	0.003008
810.6	35.69202672	32.6992	30.1695	28.0031	26.127	29.26729385	-0.088050079	0.003008
1013.25	44.61503341	40.874	37.7119	35.0039	32.6587	36.58411731	-0.110062599	0.003008

Figure 3: Calculated values of $\left(\frac{\partial \rho}{\partial T} \right)_P$ and β for Nitrogen at $T = 60^\circ\text{C}$.

1.6 Thermodynamic Derivation of H and S Relations at Constant Temperature

Derivation of the Enthalpy Relation

The starting point for the enthalpy differential is the thermodynamic identity:

$$dH = T dS + V dP. \quad (5)$$

At constant temperature, the total differential reduces to a partial dependence on pressure, expressed as:

$$\left(\frac{\partial H}{\partial P}\right)_T = T \left(\frac{\partial S}{\partial P}\right)_T + V \quad (6)$$

To relate the pressure dependence of entropy, the Maxwell relation is used from the Gibbs free energy differential,

$$dG = -S dT + V dP, \quad (7)$$

which yields the reciprocity condition:

$$\left(\frac{\partial S}{\partial P}\right)_T = - \left(\frac{\partial V}{\partial T}\right)_P \quad (8)$$

Combining these thermodynamic identities provides a direct relationship between enthalpy and volume derivatives:

$$\left(\frac{\partial H}{\partial P}\right)_T = V - T \left(\frac{\partial V}{\partial T}\right)_P \quad (9)$$

The volumetric thermal expansion coefficient, β , characterizes the relative change in volume with temperature at constant pressure:

$$\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_P \quad (10)$$

Rewriting the temperature derivative of volume in terms of β gives:

$$\left(\frac{\partial V}{\partial T}\right)_P = \beta V \quad (11)$$

Substituting this definition into the expression for the pressure derivative of enthalpy yields:

$$\left(\frac{\partial H}{\partial P}\right)_T = V(1 - \beta T) \quad (12)$$

Integrating this differential relation at constant temperature from $P = 0$ to the desired pressure P yields:

$$H(T, P) - H(T, 0) = \int_0^P V(1 - \beta T) dP \quad (13)$$

$$H(T, 0) = H^{IG}(T) \quad (14)$$

which leads to the general form:

$$\boxed{H = H^{IG} + \int_0^P V(1 - \beta T) dP} \quad (15)$$

The first term, H^{IG} , corresponds to the ideal gas enthalpy at the same temperature. It was evaluated directly in Excel using the function =HIG(N2, T_C), which returns the ideal-gas molar enthalpy of Nitrogen at the specified temperature ($T = 60^\circ\text{C}$).

The second term accounts for real-gas deviations via the pressure-dependent integral, evaluated numerically in Excel using Akima spline integration:

$$=\text{fAkima}(\text{P_kPa_range}, \text{V}*(1 - \text{beta}*T)_\text{range}, 5, 0_kPa, 70000_kPa)$$

For Nitrogen at $T = 60^\circ\text{C}$ and $P = 700$ bar, the ideal-gas enthalpy was obtained in Excel as

$$H^{IG} = 1021.02 \text{ kJ/kmol.}$$

The integral term $\int_0^P V(1 - \beta T) dP$ was evaluated using the Akima spline as

$$\int_0^P V(1 - \beta T) dP = -47.94 \text{ kJ/kmol.}$$

Hence, the molar enthalpy becomes

$$H = 973.08 \text{ kJ/kmol} = 0.973 \text{ kJ/mol.}$$

1.7 Derivation of the Entropy Relation

The same thermodynamic identities form the derivation of the entropy relation. Beginning with the enthalpy differential,

$$dH = T dS + V dP, \quad (16)$$

and the Gibbs free energy differential,

$$dG = -S dT + V dP, \quad (17)$$

the Maxwell relation follows directly:

$$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P. \quad (18)$$

The thermal expansion coefficient β can be expressed as:

$$\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_P \Rightarrow \left(\frac{\partial V}{\partial T}\right)_P = \beta V. \quad (19)$$

Therefore, the entropy variation with pressure at constant temperature can be written as:

$$\left(\frac{\partial S}{\partial P}\right)_T = -\beta V. \quad (20)$$

Integrating this expression from $P = 0$ to P leads to:

$$S(T, P) - S(T, 0) = -\int_0^P \beta V dP, \quad (21)$$

$$S(T, 0) = S^{IG}(T), \quad (22)$$

which gives the expression for entropy:

$$S = S^{IG} - \int_0^P \beta V dP \quad (23)$$

The first term, S^{IG} , corresponds to the ideal gas entropy. It was evaluated directly in Excel using the function =SIG(N2, T_C, P_bar), which returns the ideal-gas molar entropy of Nitrogen at the specified temperature ($T = 60^\circ\text{C}$) and pressure ($P = 700\text{bar}$) The integral quantifies the deviation from ideal behavior and is evaluated numerically using the Akima spline function in Excel:

$$=fAkima(P_kPa_range, \beta V_range, 5, 0_kPa, 70000_kPa)$$

The ideal-gas entropy was obtained as

$$S^{IG} = -51.12 \text{ kJ}/(\text{kmol}\cdot\text{K}).$$

The integral term $\int_0^P \beta V dP$ was computed using the Akima spline as

$$\int_0^P \beta V dP = 63.39 \text{ kJ}/(\text{kmol}\cdot\text{K}).$$

Thus, the molar entropy at $T = 60^\circ\text{C}$ and $P = 700 \text{ bar}$ is

$$S = -114.51 \text{ kJ}/(\text{kmol}\cdot\text{K}).$$

1.8 Final Computed Results

Temperature, C						T.C	T.K				
P, bar	0	25	50	75	100	Density, kmol/m3	60	333.15	beta, 1/K	V, m3/kmol	V(1-betaT), m3/kmol
0	0	0	0	0	0	0	0	0	0	0	0
1.01325	0.044615033	0.04087	0.03771	0.035	0.03266	0.036584117	-0.000110063	0.003008	27.334266	-0.062191909	101.325
3.03975	0.1338451	0.12262	0.11314	0.10501	0.09798	0.109752352	-0.000330188	0.003008	9.1114221	-0.020730636	303.975
5.06625	0.223075167	0.20437	0.18856	0.17502	0.16329	0.182920587	-0.000550313	0.003008	5.4668532	-0.012438382	506.625
10.1325	0.446150334	0.40874	0.37712	0.35004	0.32659	0.365841173	-0.001100626	0.003008	2.7334266	-0.006219191	1013.25
30.3975	1.338451002	1.22622	1.13136	1.05012	0.97976	1.097523519	-0.003301878	0.003008	0.9111422	-0.002073064	3039.75
50.6625	2.23075167	2.0437	1.88559	1.75019	1.63294	1.829205865	-0.00550313	0.003008	0.5466853	-0.001243838	5066.25
101.325	4.461503341	4.0874	3.77119	3.50039	3.26587	3.658411731	-0.01100626	0.003008	0.2733427	-0.000621919	10132.5
202.65	8.923006681	8.17481	7.54238	7.00077	6.53174	7.316823461	-0.02201252	0.003008	0.1366713	-0.00031096	20265
303.975	13.38451002	12.2622	11.3136	10.5012	9.79761	10.97523519	-0.03301878	0.003008	0.0911142	-0.000207306	30397.5
405.3	17.84601336	16.3496	15.0848	14.0015	13.0635	14.63364692	-0.04402504	0.003008	0.0683357	-0.00015548	40530
607.95	26.76902004	24.5244	22.6271	21.0023	19.5952	21.95047038	-0.066037559	0.003008	0.0455571	-0.000103653	60795
810.6	35.69202672	32.6992	30.1695	28.0031	26.127	29.26729385	-0.088050079	0.003008	0.0341678	-7.77399E-05	81060
1013.25	44.61503341	40.874	37.7119	35.0039	32.6587	36.58411731	-0.110062599	0.003008	0.0273343	-6.21919E-05	101325

Density Table at P=700bar:					
P, bar	0	25	50	75	100
700	30.82213016	28.2377	26.0531	24.1823	22.5621

(drho/dT), kmol/m3	-0.076036338
beta, 1/K	0.00300848
H(G), kJ/kmol	1021.017626
V, m3/kmol	0.03956635
V(1-betaT), m3/km	-9.00228E-05
Integral, kJ/kmol	-47.94054275
Enthalpy, kJ/kmol	973.0770831
Enthalpy, kJ/mol	0.973077083
S(G), kJ/kmolK	-51.12129273
Integral, kJ/kmolK	63.9042496
Entropy, kJ/kmolK	-114.5117177

Figure 4: Final computed thermodynamic properties of nitrogen at $T = 60^\circ\text{C}$ and $P = 700 \text{ bar}$.

2 Problem B: Statement

In this problem we are working with venting of ethylene gas from a carbon steel tank through a narrow pipe to analyze how the gas will be behaving while leaving the tank through the pipe.

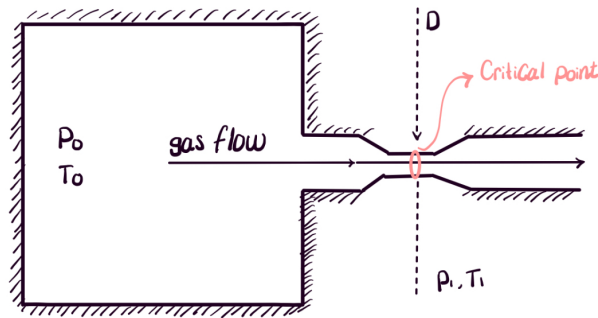


Figure 5: Schematic.

2.1 Data and Assumptions

- **Given Conditions:**

- Pressure, $P = 200$ bar
- Temperature, $T = 400$ K
- Volume, $V = 20$ m³

- **Critical Condition:**

- The critical mass flow rate corresponds to the speed of sound at the critical condition:
 $\dot{m}_{\text{critical}} = 5000$ kg/hr

- **Assumptions:**

1. The venting process is reversible and adiabatic.
2. Ethylene behaves as an ideal gas.
3. The analysis begins with the unsteady-state mass, energy, and entropy balances.
4. The kinetic energy of the gas at the narrow section of the pipe cannot be ignored.

2.2 Calculation Procedure

Start by performing the calculations at **steady state** (S_0).

Next, add **ethylene** as a component in the database (**comp2**).

Then, set up a table for S_0 and S_1 :

Table 1: Summary of System States

Property	S_0 (Initial)	S_1 (Critical)
Pressure (bar)	200	—
Temperature (K)	400	—
Volume (m ³)	20	20
Mass (kg)	—	—

To calculate the **ideal gas heat capacities** at a specified temperature for a given component (in kJ/kmol·K), use:

$$C_p = \text{CpIG}(\text{"comp2"}, T(\text{C})) = 54.20603068 \text{ kJ}/(\text{kmol}\cdot\text{K})$$

$$C_v = \text{CvIG}(\text{"comp2"}, T(\text{C})) = 45.89156806 \text{ kJ}/(\text{kmol}\cdot\text{K})$$

Also calculate the **gas constant** R in kJ/(kmol·K), using the relation:

$$R = C_p - C_v$$

Substituting the values:

$$R = 61.3 \text{ kJ}/(\text{kmol}\cdot\text{K}) - 53.0 \text{ kJ}/(\text{kmol}\cdot\text{K}) = 8.314462618 \text{ kJ}/(\text{kmol}\cdot\text{K})$$

Calculate the **heat capacity ratio** (γ) in kJ/(kmol·K), which is defined as:

$$\gamma = \frac{C_p}{C_v}$$

Substituting the values:

$$\gamma = \frac{61.3 \text{ kJ}/(\text{kmol}\cdot\text{K})}{53.0 \text{ kJ}/(\text{kmol}\cdot\text{K})} = 1.181176259$$

To calculate the **ideal gas enthalpy** and **entropy** at a specified temperature and pressure for a given component (using units of kJ/kmol and kJ/kmol·K), use:

$$H = \text{HIG}(\text{"comp2"}, T(\text{C}), P(\text{bar})) = 57323.97357 \text{ kJ}/\text{kmol}$$

$$S = \text{SIG}(\text{"comp2"}, T(\text{C}), P(\text{bar})) = -82.7063037 \text{ kJ}/(\text{kmol}\cdot\text{K})$$

Now begin performing the calculations for S_1 , which represents the **critical condition** of the system.

At this stage, the gas velocity reaches the speed of sound in the narrow section of the pipe. Determine the corresponding pressure, temperature, and mass of ethylene using the relationships defined for critical (choked) flow.

For the calculations at S_1 , begin by **assuming** two initial guess values for the temperature and pressure. These guessed values are:

$$T = 90 \text{ }^\circ\text{C} \quad \text{and} \quad P = 110 \text{ bar}$$

The **volume of the system** will remain the same as in S_0 .

Using these guessed values, **apply all the formulas previously calculated for S_0** (such as those for C_p , C_v , R , γ , H , and S) to determine the corresponding thermodynamic properties at S_1 based on our assumptions.

Calculate the mass flow rate in kg/s.

Using $\dot{m}_{\text{critical}} = 5000 \text{ kg/hr}$

The mass flow rate in seconds is calculated as:

$$\dot{m} = \frac{5000 \text{ kg/hr}}{3600 \text{ s/hr}} = 1.388888889 \text{ kg/s} \quad (24)$$

Calculate the molar flow rate in kmol/s, which is given by the mass flow rate divided by the molecular weight of ethylene.

$$\dot{n} = \frac{\dot{m}}{MW_{\text{C}_2\text{H}_4}} = \frac{1.388888889 \text{ kg/s}}{28.05 \text{ kg/kmol}} = 0.049507695 \text{ kmol/s} \quad (25)$$

Then, calculate the speed of sound in m/s.

It is given by the square root of $\gamma RT(K)$ divided by $\left(\frac{MW_{\text{C}_2\text{H}_4}}{100}\right)$.

$$a = \sqrt{\frac{\gamma R T(K)}{MW_{\text{C}_2\text{H}_4}/1000}} \text{ m/s} \quad (26)$$

Now calculate the kinetic energy in J/kg.

It is given by one half of the square of the speed of sound.

$$KE = \frac{1}{2} a^2 \text{ J/kg} \quad (27)$$

Convert the kinetic energy to kJ/kmol using the molecular weight of ethylene.

It is calculated as the specific kinetic energy divided by 1000 and multiplied by $MW_{C_2H_4}$.

$$KE_{\text{kJ/kmol}} = \frac{KE}{1000} \times MW_{C_2H_4} \quad (28)$$

Perform the enthalpy and entropy balances.

The enthalpy balance is given by:

$$H = H_{\text{in}} - H_{\text{out}} - KE \text{ (kJ/kmol)} \quad (29)$$

The entropy balance is given by:

$$S = S_{\text{in}} - S_{\text{out}} \quad (30)$$

Calculate the density in kmol/m³ using the density function:

$$\rho = \text{density}(T(C), p(\text{Bar})) \text{ kmol/m}^3 \quad (31)$$

Using the calculated density, determine the cross-sectional area in m² as:

$$A = \frac{\dot{n}}{a \rho} \text{ m}^2 \quad (32)$$

Calculate the inner diameter in mm using the cross-sectional area.

$$D = 2 \times \sqrt{\frac{A}{\pi}} \times 1000$$

Finally, perform a goal-seek to satisfy the balances.

Adjust the temperature to drive the enthalpy balance to zero:

$$H = H_{\text{in}} - H_{\text{out}} - KE = 0 \quad (34)$$

Similarly, adjust the pressure to drive the entropy balance to zero:

$$S = S_{\text{in}} - S_{\text{out}} = 0 \quad (35)$$

2.3 Data Table

	S0	S1
T,K	400	365.383
T,C	126.85	92.2329
P,bar	200	112.984
V,m3	20	20
Molarflow,kmol,s		0.04951
Mass flow,kg,s		1.38889
Cp, kJ/kmol k	54.206	50.7303
Cv,kJ/kmol k	45.8916	42.4158
R,kJ/kmol k	8.31446	8.31446
γ	1.18118	1.19602
Enthalpy,kJ/kmol	57324	55507.2
Entropy,kJ/kmol	-82.7063	-82.7063
Speed of sound, m/s		359.885
KE,J/kg		64758.5
KE,kJ/kmol		1816.74
Enthalpy balance		5E-10
Entropy Balance		-1.7E-09
Area,m2		3.7E-05
D, mm		6.86265
Density,kmol/m^3		3.71908

Figure 6: Data

2.4 Thermodynamic Derivations

Specific enthalpy is defined as

$$h = u + p v, \quad (36)$$

where u is the specific internal energy, p is pressure, and v is specific volume ($v = 1/\rho$).

Differentiate (36):

$$dh = du + v dp + p dv. \quad (37)$$

2. Use the Gibbs relation

From the thermodynamic (Gibbs) identity for a simple compressible substance,

$$du = T ds - p dv, \quad (38)$$

where T is temperature and s is specific entropy. Substitute (38) into (37):

$$\begin{aligned} dh &= (T ds - p dv) + v dp + p dv \\ &= T ds + v dp. \end{aligned} \quad (39)$$

3. Isentropic assumption

Under the isentropic assumption $ds = 0$, (39) reduces to

$$dh = v dp. \quad (40)$$

Integrating formally with respect to p gives

$$h(p) = \int v(p) dp + C, \quad (41)$$

where C is an integration constant. (Integration by parts shows $\int v dp = pv - \int p dv$, so $h = pv$ only under specific choices of reference or when the remaining integral is absorbed into the constant.)

4. Steady-flow energy (Bernoulli-style)

For steady adiabatic flow with no shaft work, the steady-flow energy equation per unit mass may be written in Bernoulli-like form:

$$h + \frac{1}{2}V^2 + gz = \text{constant along a streamline.} \quad (42)$$

Here V is the flow speed at the point considered, g gravitational acceleration, and z elevation.

5. Neglect potential energy ($gz = 0$)

If potential-energy changes are negligible (for example, small elevation change or horizontal flow), set $gz \approx 0$ in (42). Define the stagnation (total) enthalpy H_0 as the constant value:

$$H_0 \equiv h + \frac{1}{2}V^2. \quad (\text{with } gz = 0) \quad (43)$$

Because the right-hand side of (42) is constant along the streamline under the stated assumptions, H_0 is constant along the streamline.

6. Two-point relation

Evaluating the constant H_0 at two points along the same streamline (point 0: stagnation point, and point 1: flow point) gives

$$H_0 = h_1 + \frac{1}{2}V_1^2. \quad (44)$$

This is the standard relationship used when converting kinetic energy into enthalpy (or vice versa) under the given assumptions.

2.5 Calculating Initial Mass of Ethylene in the Tank

calculating the initial mass of ethylene in the tank.

The mass is calculated using the ideal gas law (without unit conversions):

$$m = \frac{P V}{R T} \cdot MW_{C_2H_4} \text{ kg} \quad (45)$$

Including unit conversions to use P in bar, V in m^3 , T in K, and MW in kg/kmol:

$$m = \frac{P V 10^6}{83.144 T} \cdot \frac{MW_{C_2H_4}}{1000} = kg \quad (46)$$

Where:

- 10^6 converts $\text{bar} \cdot m^3$ to $\text{Pa} \cdot m^3$
- 83.144 is the gas constant R in $\text{bar} \cdot m^3 / \text{kmol} \cdot K$
- $MW/1000$ converts molar mass from kg/kmol to kg/mol to match units

Using the given values for pressure, volume, and temperature, the initial mass of ethylene in the tank is calculated as:

$$m_{\text{initial}} = 3374.15 \text{ kg} \quad (47)$$

2.6 Calculations of Temperature and Pressure at Critical Mass Flow

Pressure, temperature, and diameter were calculated at the narrow section of the venting pipe. The values are:

- Temperature: 92.23285992 °C
- Temperature: 365.3828599 K
- Pressure: 112.9842983 bar
- Diameter: 6.862649343 mm

2.7 Mass of Ethylene in the Tank at Critical Condition

In order to solve for the mass at the critical conditions, we will use the same equation that was applied to calculate the initial mass of ethylene in the tank. However, this time we will substitute the temperature and pressure corresponding to State 1 (critical conditions):

$$T = 365.3828599 \text{ K} \quad \text{and} \quad P = 112.9842987 \text{ bar}$$

Using these values in the mass equation, the mass of ethylene at the critical point is calculated to be:

$$m = 2086.718169 \text{ kg}$$

2.8 Mass of Ethylene in the Tank at 2300 psig

Calculate the mass of ethylene in the tank when the pressure gauge reads 2300 psig. This corresponds to an absolute pressure of:

- First, convert the pressure from **psia** to **bar** using the **psiatobar** function:

$$P = 2300 \text{ psia} = 158.578865 \text{ bar}$$

- Next, we will assume an initial (guessed) temperature:

$$T = 100 \text{ }^{\circ}\text{C}$$

- Using this guessed temperature, we will drag the Excel formulas downward to calculate the following properties:

- Molar and mass flow rates (kmol/s and kg/s)
- Specific heat capacities C_p and C_v (kJ/kmol·K)
- Gas constant R (kJ/kmol·K)
- Ratio of specific heats $\gamma = \frac{C_p}{C_v}$
- Enthalpy and entropy (kJ/kmol)
- Speed of sound (m/s)
- Kinetic energy (J/kg)

- Then, we perform an **entropy balance** between State 1 and State 2:

$$s = s_{\text{in}} - s_{\text{out}}$$

- We will use the **Goal Seek** function in Excel to adjust the temperature until the entropy balance equals zero.
- The converged temperature at equilibrium is found to be:

$$T = 112.6813444 \text{ }^{\circ}\text{C}$$

- Finally, we can convert this temperature to Kelvin using the **CtoK** function:

$$T = 385.6813444 \text{ K}$$

To calculate the mass of ethylene in the tank at a pressure of 2300 psia, we will use the **same equation** that was previously applied to calculate the initial and critical masses. However, for this case, we will substitute the temperature and pressure corresponding to **State 2 (S2)**.

$$T = 112.6813444 \text{ }^{\circ}\text{C} \quad (385.6813444 \text{ K})$$

$$P = 158.578865 \text{ bar}$$

The general mass equation is given by:

$$m = \frac{P \times V \times 10^6}{R \times T} \times \frac{\text{MOLW2}}{1000}$$

where:

- P = Pressure (bar)
- V = Volume (m^3)
- R = Universal gas constant ($\text{kJ/kmol}\cdot\text{K}$)
- T = Temperature (K)
- MOLW2 = Molar mass of ethylene (kg/kmol)
- 10^6 = Conversion from kJ to J
- 1000 = Conversion from J/kg to kJ/kmol

By substituting the **State 2** values of temperature and pressure into this equation, the mass of ethylene in the tank is calculated as:

$m = 2773.586233 \text{ kg}$

3 Conclusion

This report showed how thermodynamics can be applied to both real and ideal gases using data, equations, and simple numerical tools. For the Nitrogen problem, real-gas behavior was studied using Akima spline interpolation to find density and its temperature change at high pressure. From this, the thermal expansion coefficient, molar enthalpy, and entropy were calculated. The results showed that nitrogen behaves differently from an ideal gas at high pressures, and that careful interpolation helps make the results smoother and more reliable.

The ethylene venting problem focused on an ideal-gas process that was both adiabatic and reversible. By applying the conservation of mass, energy, and entropy, the analysis explained how gas escapes the tank and reaches the critical condition at the narrow part of the pipe. The problem also showed why kinetic energy must be included when gas flow speeds become very high.

Overall, the work helped connect theory with practical problem-solving. It showed how thermodynamic formulas, data fitting, and energy balances can be combined to understand how gases behave in real engineering systems.